

Computational Study on Vaporizing Liquid Micro-Thruster

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INTRODUCTION

The development of mini-satellite has been paid much attention recently due to the advantages of lower cost and technical threshold. In contrast to traditional satellites, the mini-satellites needed for the space missions of new generation require micro-propulsion units [1] with an extremely low level of thrust and precise spatial maneuverings. Therefore, various MEMS-based micro-thrusters have been proposed by governmental, industrial and academic research groups to fulfill those specific requirements in recent years [2-9]. However, the gas micro-thruster system requires excessive fuel masses which demand the high-pressure-gas tank and supply line. On the other hand, the catalyzed micro-chemical thrusters have the problems of the damage of the catalyst chamber in the heterogeneously catalyzed micro-chemical thruster [3] and line deposition of catalyst in the homogeneously catalyzed micro-chemical thruster. In addition, the electric-based micro-propulsion requires too much power consumption to be offered for the micro/nano satellites. Thus, the vaporizing liquid micro-thruster (VLM) based on the solar thermal energy is proposed to avoid the drawbacks described above. Furthermore, this kind of VLM holds the advantages of the simplicity, high reliability, prevention of fuel or catalyst aging problem, environmentally friendly nature and inexhaustibility of the solar energy in space. Therefore, the research of solar vaporizing liquid micro-thruster possesses the development of great potential.

Based on our previous study [10], the straight heating channel of the VLM still fills with the tiny droplets, even with suitable operation conditions, as shown in Fig. 1. To improve the propellant efficiency and then the performance of the VLM, the vaporization rate should be significantly enhanced. Due to the consideration of the cost and time saving, numerical simulation was employed to examine the effects of the heating channel configuration of the VLM during the design phase. A simplified 3-D computational model for the vaporizing liquid micro-thruster was developed in the present study due to the complex phenomena involved the liquid phase, vapor phase, evaporation, capillary force, heat transfer, and dynamic instability. The designs of the heating channel of the VLM were estimated by the trajectories of the seeding droplets which represent the tiny droplets described above. All computations are performed by the commercial computational fluid dynamics (CFD) software from ESI Group [11]

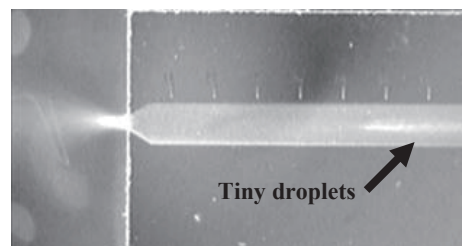


FIGURE 1. The tiny droplets in micro-thruster with the outlet width of 400 μm at wall temperature= 300 $^{\circ}\text{C}$ and mass flow rate=0.25 cc/min.

PROBLEM DESCRIPTION AND COMPUTATIONAL MODELS

The flow field in vaporizing liquid micro-thruster resembles the two-phase flow phenomena in micro-channels and is associated with liquid phase, vapor phase, evaporation, capillary force, heat transfer, and dynamic instability [12-15]. The phenomena are too complicated to model the problem completely in the current numerical studies. Therefore, a simplified 3-D computational mesh for the vaporizing liquid micro-thruster was developed in the present study.

The hypotheses in the numerical simulations were described as follows. The flow field in the VLM included the continuous phase and the discrete phase. The continuous phase was the gaseous water which flowed into the VLM in the state of pure saturated vapor. The discrete phase was the tiny water droplets without evaporation. Based on the realistic structure of the VLM and experimental conditions [10], the upper surface of the simulation model was assumed as an adiabatic wall and the rest of surfaces as isothermal walls due to the extremely low thermal conductivity of Pyrex glass and high thermal conductivity of silicon. The wall temperature was set to be 300 $^{\circ}\text{C}$. The inlet condition was assumed as a constant mass flow rate and the atmospheric pressure was 1 bar. Ideal gas law was adopted to compute the properties of fluid. The flow was laminar and assumed to follow the no-slip boundary condition. The flow states were transient. The numerical solutions resulted from solving the conservation equations of mass, momentum, and energy. The discrete droplets were tracked through the computational domain by solving the Lagrangian equations. The normal and parallel coefficients of restitution were set as 0.3 and 0.9 respectively by referring to the data from Yu et al. [16] and our experimental results [10]. The size of the seeding droplets was set as 15 μm , which was a

half of the minimum identifiable droplet diameter in one pixel, to represent the invisible droplet in the heating channel. The gas mass flow rates were 2.08×10^{-6} kg/s (water evaporation rate is about 0.125 cc/min at flow upstream). The gas mass flow rate was selected based on the assumption that the evaporation ratio was 50 percent at the 0.25 cc/min of liquid water. The basic requirements for the pattern design in the VLM were to block the droplets stream from direct discharge and to increase the droplet collisions with the high temperature wall. Also, the streamline contours should be considered to prevent high pressures from being generated in the VLM and then diminish the difficulty in the propellant injection. The geometric patterns and main dimensions of the VLM used for computation were exhibited in Figs. 2(I) to 2(V) and the depth was 100 μm . In model V, four pillar spacing conditions were computed. The center-to-center distances between two lines of pillar were 250 μm , 300 μm , 350 μm , and 400 μm , respectively.

RESULTS AND DISCUSSION

Simulations were performed for all different computational models as described above. The simulation results of micro-thruster with blank channel showed that the droplets seeded from the inlet directly flowed out the VLM except hitting the beveled side of the nozzle, as showed in Fig. 3. The droplet stream lightly expanded at downstream and squeezed through the nozzle. The trajectories of the droplets were symmetric with center line of the VLM. The flying duration of droplet stream ejected from the middle of the inlet was about 0.35 milliseconds and that from the outside of the inlet was about 0.42 milliseconds in the VLM. Figures 4(a) to 4(c) showed the trajectories of the droplets from model II. The droplets experienced 12 to 16 times of collisions with the wall before flowed into the nozzle. The flying durations of the droplets were about 1.11 milliseconds to 1.45 milliseconds. From the viewpoint of heat transfer, the model II was a suitable design. However, the droplet stream leaned to one side as they fled into the nozzle. This phenomenon caused thrust force to deflect. The design of the model II failed in the requirement of maintaining well axial thrust. The droplets in model III experienced 3 or 5 times of collisions with the wall before flowed into the nozzle, as showed in Figs. 5(a) to 5(c). The flying durations of the droplets were about 0.48 milliseconds to 1 millisecond. The trajectories of the droplets in model III were symmetric with center line of the VLM. Compared with model IV, the design of the model III was not good enough. The model IV displayed that the droplets experienced 6 or 7 times of collisions with the wall before flowed into the nozzle, as showed in Figs. 6(a) to 6(c). The flying durations of the droplets were about 0.58 milliseconds to 0.79 milliseconds. Nevertheless, the second semi-circle block in model IV was not hit by droplet and was superfluous. Additionally, the droplets could hit the tip of the second bevel fin, as shown in Fig. 6(b). This condition confined the droplet stream to the middle area of the heating channel and made the behind fin, if added, inactive. Increasing fin, if necessary, was not able to guarantee the more collisions. The design of the pillar array could be the best solution. Figures 7(a) to 7(c) displayed the results of the design with pillar spacing of 400 μm (center-to-center distances between two lines of pillar). The droplet stream ejected from the middle of the inlet fled through the pillar array after 5 times of collisions and collided the wall again before flowed into the nozzle, as shown in Fig. 7(a). The flying durations of the droplets were about 0.76 milliseconds. The droplet stream deviated from the middle of the inlet had 9 times of collisions and the flying durations of the droplets were about 0.97 milliseconds, as shown in Fig. 7(b). The 6 to 7 times of collisions happened to the outside droplet stream and the flying durations of the

droplets were about 0.76 milliseconds, as shown in Fig. 7(c). Also, the pillar array dispersed the droplets visibly. As the pillar spacing reduced, the time of collision and the flying duration of the droplet rose. At the design with pillar spacing of 250 μm , the collisions increased to 7 to 14 times, as shown in Figs. 8(a) to 8(b), and the flying durations of the droplets grew to the range of the 0.74 milliseconds to 0.99 milliseconds. Comparing with the model IV, the temperatures of the droplets flowing into the nozzle in the model V were higher than that in model IV, as displayed in Figs. 9(a) and 9(b).

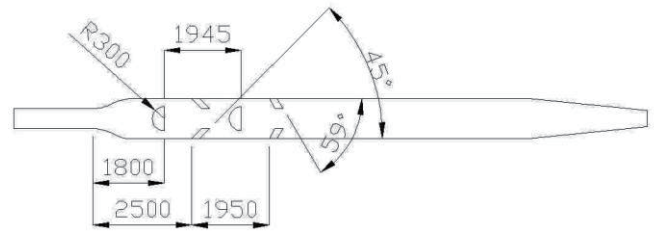
CONCLUSION

Simulation results showed that the models which consist of the pillar array structure induced more droplet collisions than those models constructed by the circle block, semi-circle block, ellipsoid block, and bevel fins, and avoided the deflection of the droplets which was revealed in the serpentine model. In addition, within the VLM chip, the wall temperature was much higher than the fluid temperature and certainly higher than those of the droplets in the flow due to the saturation temperature of fluid. Therefore, the more collisions occurred, the higher the droplet temperature was, due to the high heat flux caused by conduction between the droplets and the wall. Also, the longer flying durations of the droplets caused by the collisions increased the droplet temperature. The high droplet temperature implied the increase of the evaporation rate in the real VLM chip. Furthermore, the design with the pillar array structure provided a well axial thrust due to the symmetric trajectories of the droplets. From the viewpoint of the propellant efficiency and the properties of the ejected plume, the model with the pillar array structure is the best design pattern in the present study. In addition, the optimum pillar spacing should be 250 μm (center-to-center distances between two lines of pillar) in this work.

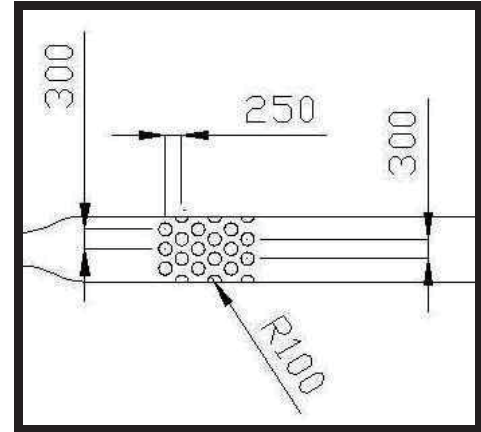
REFERENCES

- [1] J. Mueller, Thruster options for microspacecraft: a review and evaluation of existing hardware and emerging technologies, in: 33rd AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, AIAA 97-3058, Seattle, USA, 6–9 July 1997.
- [2] Köhler J, Bejhed J, Kratz H, Bruhn F, Lindberg U, Hjort K, and Stenmark L, “A hybrid cold gas microthruster system for spacecraft,” *Sensors and Actuators A*, Vol. 97-98, pp. 587–598, 2002.
- [3] Hitt D L, Zakrzewski C M, and Thomas M A, “MEMS-based satellite micropropulsion via catalyzed hydrogen peroxide decomposition,” *Smart Materials And Structures*, Vol. 10, pp. 1163–1175, 2001.
- [4] <http://www-leland.stanford.edu/group/pdl/EP/EP.html>.
- [5] Xiong J, Zhou Z, Ye X, Wang X, Feng Y and Li Y, “A colloid micro-thruster system,” *Microelectronic Engineering*, Vol. 61-62, pp. 1031–1037, 2002.
- [6] Mukerjee E V, Wallace A P, Yan K Y, Howard D W, Smith R L, Collins S D, “Vaporizing liquid microthruster,” *Sensors and Actuators A*, Vol. 83, Issues 1-3, pp. 231–236, 2000.
- [7] Ye X Y, Tang F, Ding H Q, and Zhou Z Y, “Study of a vaporizing water micro-thruster,” *Sensors and Actuators A*, Vol. 89, Issues 1-2, pp. 159–165, 2001.
- [8] Maurya D K, Das S and Lahiri S K, “An analytical model of a silicon MEMS vaporizing liquid microthruster and some experimental studies,” *Sensors and Actuators A*, Vol.122, Issue 1, pp. 159–166, 2005.

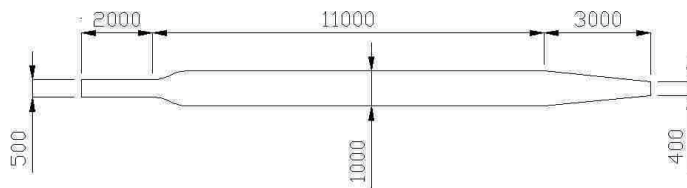
- [9] Maurya D K, Das S and Lahiri S K, "Silicon MEMS vaporizing liquid microthruster with internal microheater", *J. Micromech. Microeng.*, Vol. 15, p. 966-970, 2005.
- [10] Chia-Chin Chen, Chien-Wei Liu, Heng-Chuan Kan, Lee-Her Hu, Guey-Shin Chang, Ming-Chih Cheng, and Bau-Tong Dai, "Simulation Aid to the Research Analysis for Vaporizing Liquid Micro-Thruster," The 17th National Computational Fluid Dynamics Conference, Zhongli, July, 2010.
- [11] CFD-ACE+ 2003, ESI Group, Paris, France.
- [12] Chen C -C, Liu C -W, Kan H -C, Hu L -H, Chang G -S, Cheng M -C, and Dai B -T, "Simulation and Experiment Research on Vaporizing Liquid Micro-Thruster," *Sensors and Actuators A*, January, Vol. 157, Issue 1, pp. 140-149, 2010.
- [13] Wu H Y and Cheng P, "Liquid/two phase/vapor alternating flow during boiling microchannels at high heat flux," *International communications in heat and mass transfer*, Vol. 30, Issue 3, pp. 295-302, 2003.
- [14] Karsli S, Yilmaz M, and Comakli O, "The effect of internal surface modification on flow instabilities in forced convection boiling in a horizontal tube," *International journal of heat and fluid flow*, Vol. 23, Issue 6, pp. 776-791, 2002.
- [15] Kakac S and Bon B, "A Review of two-phase flow dynamic instabilities in tube boiling systems," *International journal of heat and mass transfer*, Vol. 51, Issues 3-4, pp. 399-433, 2008.
- [16] Yu Z, Ge Y, Fan L -S, "Multi-scale simulation of oblique collisions of a droplet on a surface in the Leidenfrost regime," *Chemical Engineering Science*, Vol. 62, pp. 3462-3472, 2007.



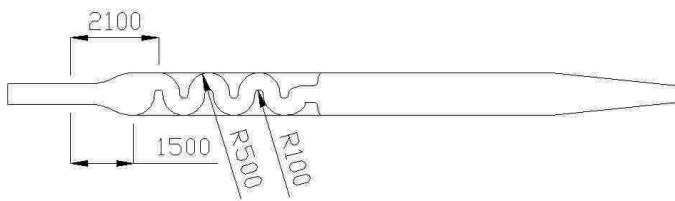
(IV)



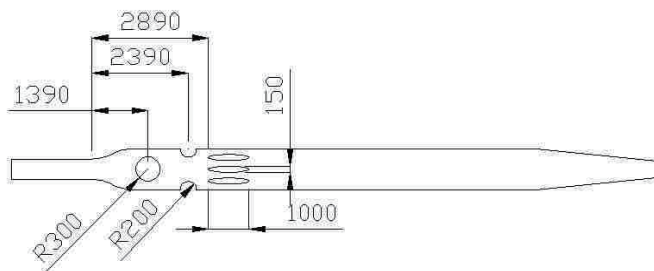
(V)



(I)



(II)



(III)

FIGURE 2 The geometric patterns of the VLM used for computation. (dimension in μm).



FIGURE 3. Simulation results from model I.

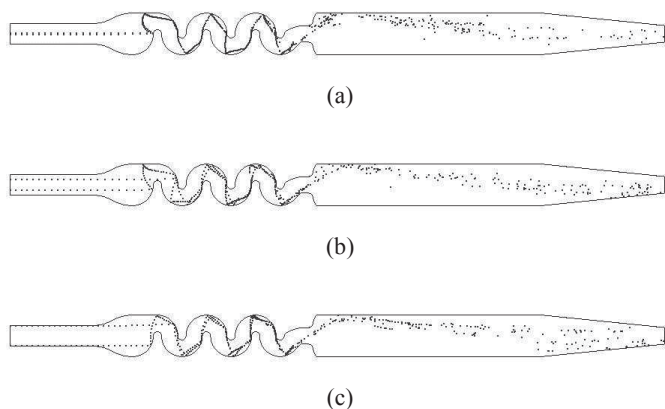


FIGURE 4. Simulation results from model II.

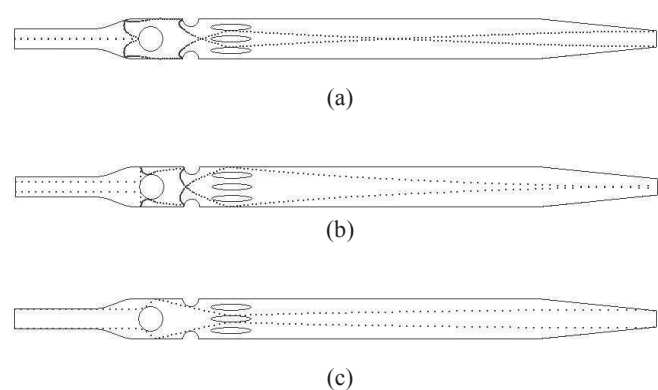


FIGURE 5. Simulation results from model III.

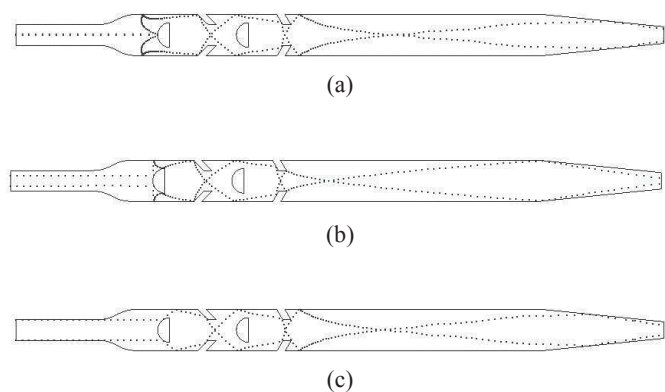


FIGURE 6. Simulation results from model IV.

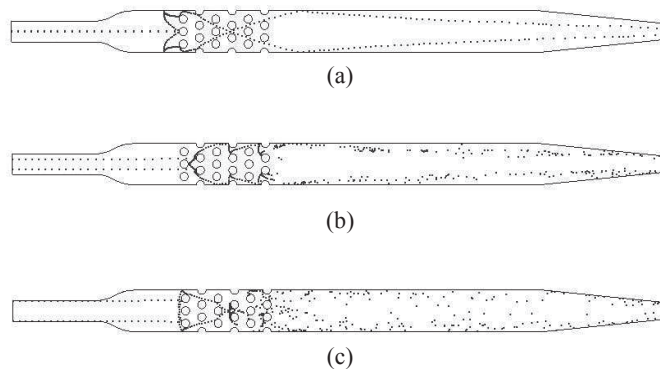


FIGURE 7. Simulation results from model V (with pillar spacing of $400\mu\text{m}$).

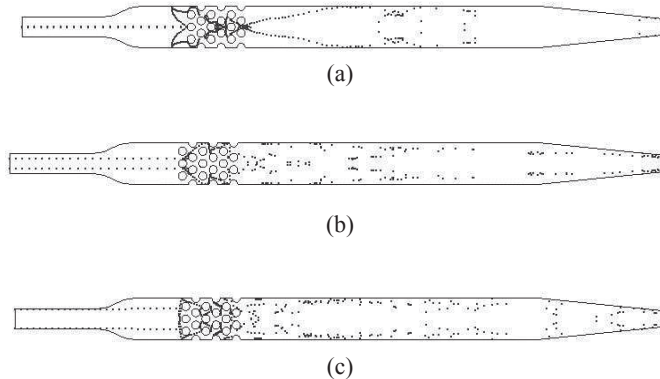


FIGURE 8. Simulation results from model V (with pillar spacing of $250\mu\text{m}$).

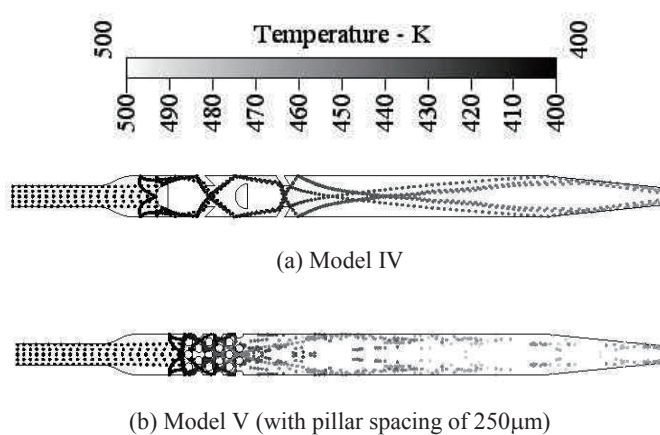


FIGURE 9. Temperature distribution of the droplets in the VLM.